Rolling-window U.S. stock returns

$1,000 lump sum

Monthly data, Aug 1926 – Mar 2026 (1,195 months)

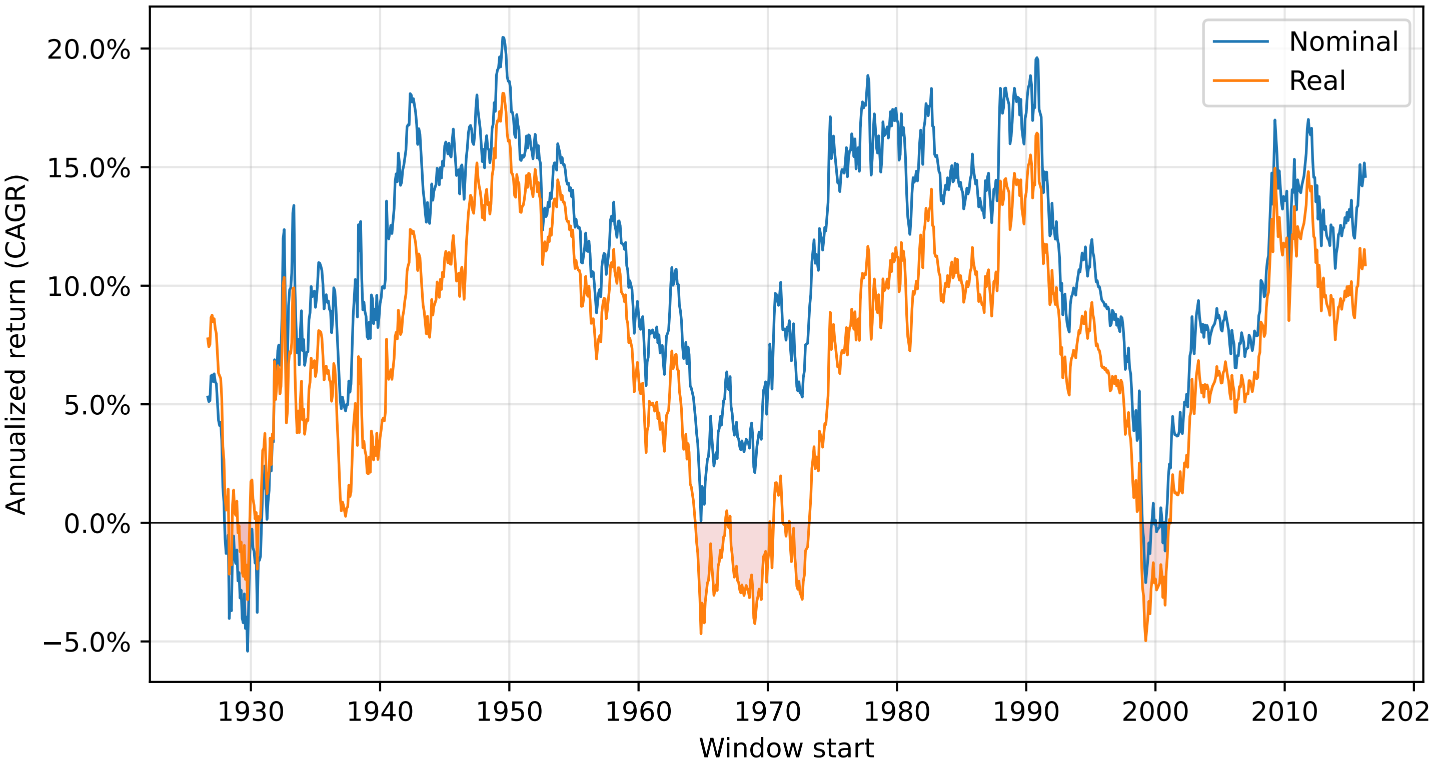
Generated 2026-05-17

Full-history compound annual growth rate (market)		
Nominal	10.20%	[1]
Real (CPI-deflated)	6.99%	[1][2]
Mean 10-year rolling CAGR		
Nominal	10.56%	[3]
Real	7.05%	[3]
Highest 10-year rolling CAGR		
Nominal	+20.48%	[3]
Real	+18.12%	[3]
Lowest 10-year rolling CAGR		
Nominal	-5.41%	[3]
Real	-4.97%	[3]
10-year rolling windows with CAGR ≤ 0		
Nominal	4.65% (50 of 1,076)	[3]
Real	12.45% (134 of 1,076)	[3]
Total 10-year rolling windows	1,076	[3]
Window labeling	by start date	[3]
Total contributed per window	$1,000	

Invest $1,000 at the start of each rolling N-year window and hold to the end. The metric is annualized CAGR — the constant annual rate that grows $1,000 to the terminal value over N years.

10-year rolling CAGR — Nominal & Real

Each point is the annualized return of a 10-year window beginning on that month. Red shading marks the negative region.

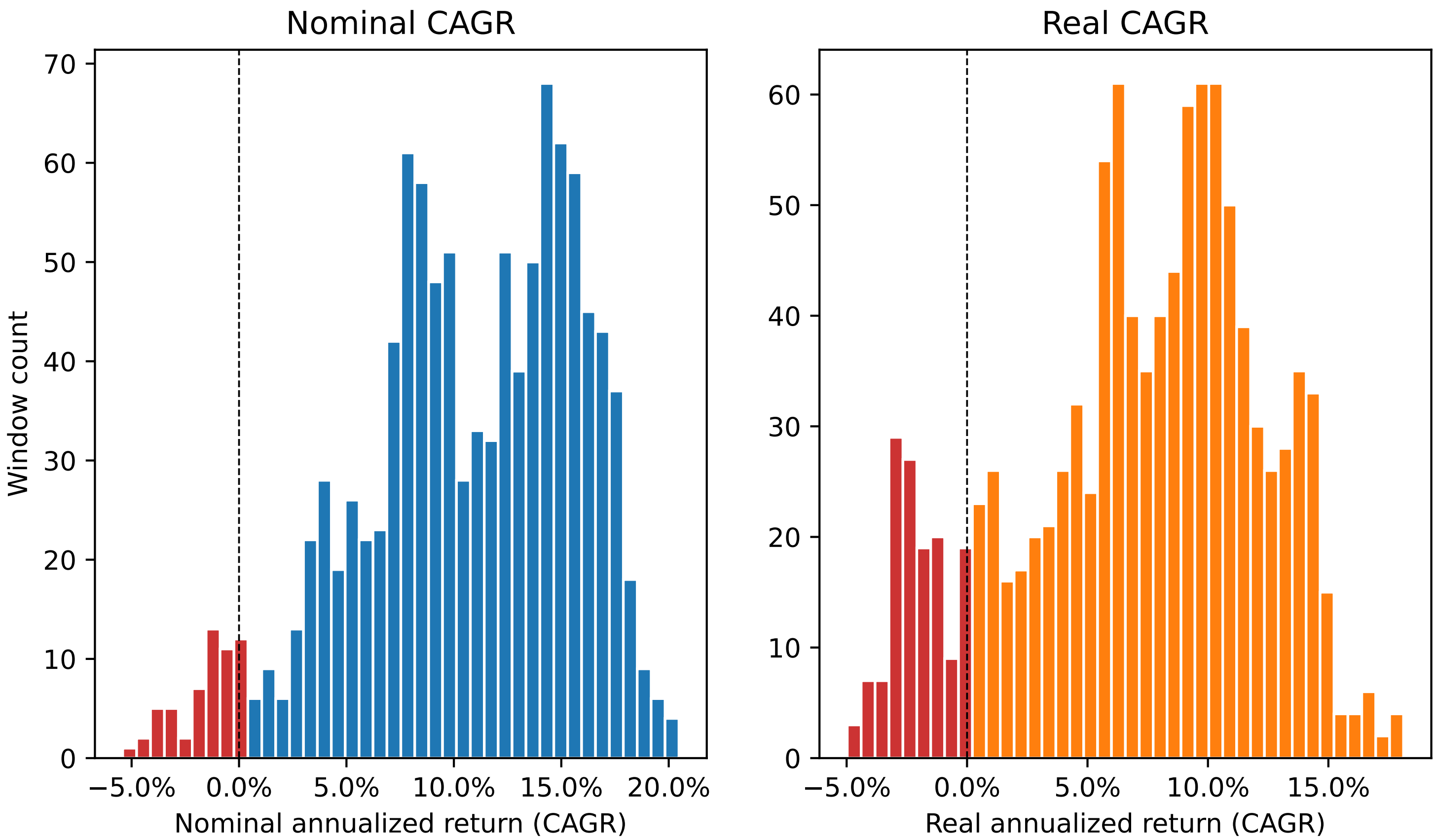


Summary statistics

	Nominal	Real
Mean	10.56%	7.05%
Median	11.10%	7.76%
Std. dev.	5.21%	5.07%
Min	-5.41%	-4.97%
Max	20.48%	18.12%
Share ≤ 0	4.65%	12.45%

Distribution of 10-year rolling CAGR — Nominal & Real

Histogram of every 10-year window in the sample. Negative bins are shown in red.



Share of windows with CAGR ≤ 0

Nominal: 4.65%   (50 of 1,076 windows)

Real: 12.45%   (134 of 1,076 windows)

"≤ 0" means an investor who followed this schedule over the full 10-year window would have ended at or below their total contributions, after compounding (and after inflation, for the real series).

[3]

Best and worst 10-year windows

Top 5 highest and lowest annualized returns (CAGR), ranked by window-start month.

Best nominal

Start	End	CAGR	$1,000 →
Jun 1949	May 1959	20.48%	$6,442
Jul 1949	Jun 1959	20.45%	$6,429
Aug 1949	Jul 1959	20.20%	$6,295
May 1949	Apr 1959	19.89%	$6,137
Sep 1949	Aug 1959	19.74%	$6,058

Worst nominal

Start	End	CAGR	$1,000 →
Sep 1929	Aug 1939	**-5.41%**	**$573**
Jul 1929	Jun 1939	**-4.45%**	**$634**
May 1929	Apr 1939	**-4.22%**	**$650**
Apr 1928	Mar 1938	**-4.03%**	**$662**
Apr 1929	Mar 1939	**-4.03%**	**$663**

Best real

Start	End	CAGR	$1,000 →
Jun 1949	May 1959	18.12%	$5,287
Jul 1949	Jun 1959	18.10%	$5,280
Aug 1949	Jul 1959	17.72%	$5,109
May 1949	Apr 1959	17.60%	$5,057
Mar 1949	Feb 1959	17.35%	$4,953

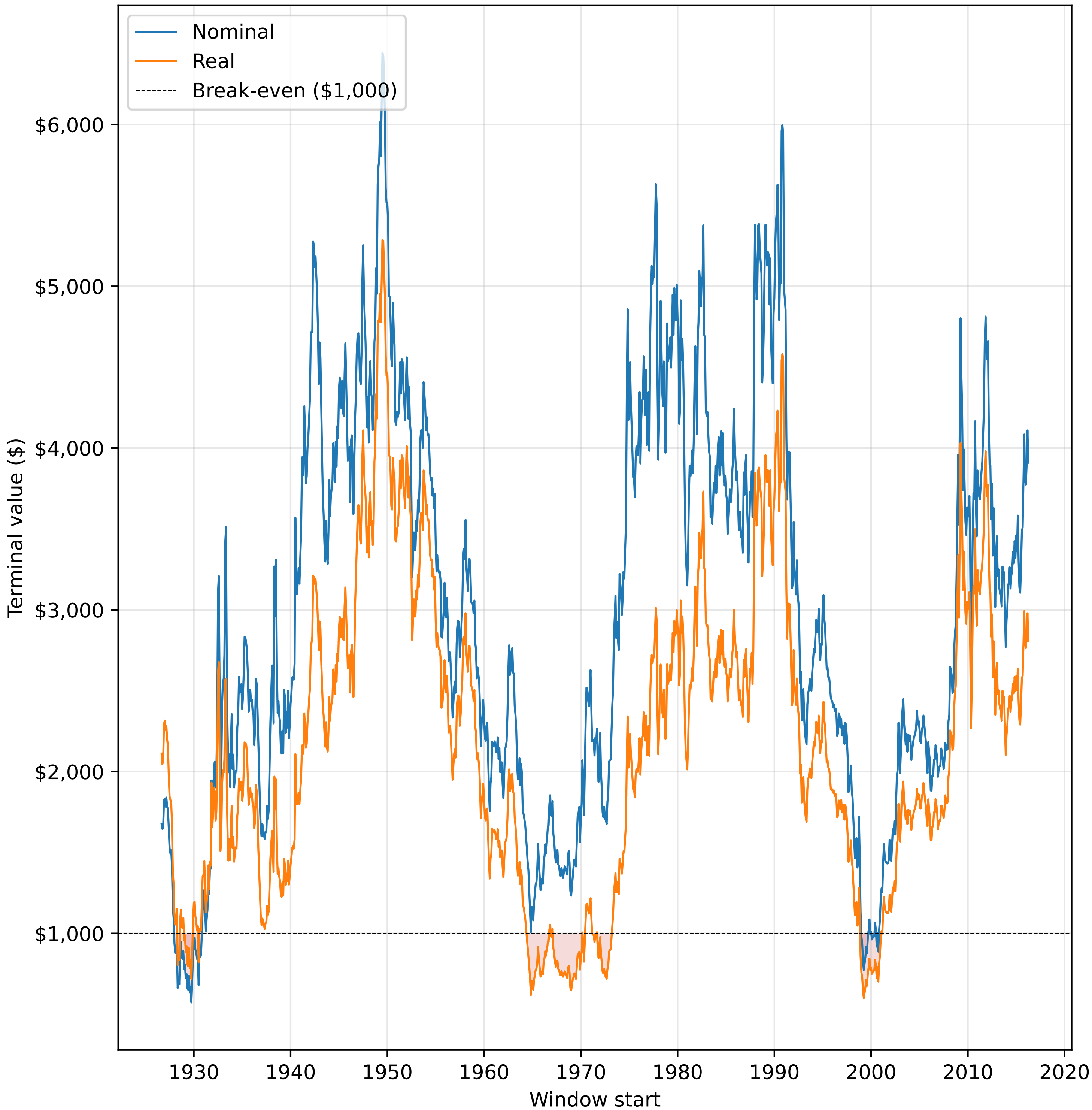
Worst real

Start	End	CAGR	$1,000 →
Mar 1999	Feb 2009	**-4.97%**	**$600**
Oct 1964	Sep 1974	**-4.68%**	**$619**
Apr 1999	Mar 2009	**-4.51%**	**$630**
Feb 1999	Jan 2009	**-4.28%**	**$646**
Dec 1968	Nov 1978	**-4.25%**	**$648**

inal value of the full contribution schedule, held to the end of the 10-year window. [4] Red marks a negative CAGR or a terminal below t

Terminal value of $1,000 after 10 years — Nominal & Real

re $1,000 contributed under this schedule would have landed, by start month. Red shading marks where the terminal fell below break-e



Sources and methodology

Data sources

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[1] Fama/French Research Data Factors, monthly file. Ken French Data Library,
    Tuck School of Business, Dartmouth.
    https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
    Columns used: Mkt-RF (market excess return), RF (risk-free rate).
    Values are published as percent (2.50 = 2.5%) and divided by 100 on ingest.

[2] U.S. CPI for All Urban Consumers, not seasonally adjusted (CPIAUCNS).
    Federal Reserve Bank of St. Louis (FRED).
    https://fred.stlouisfed.org/series/CPIAUCNS
    The not-seasonally-adjusted series is used deliberately — the SA series
    (CPIAUCSL) only begins in 1947 and would truncate 21 years of history.
```

Series construction

```
nominal_return = (Mkt-RF + RF) / 100  — total U.S. market return.
inflation      = CPI.pct_change()    — month-over-month CPI change.
real_return    = (1 + nominal) / (1 + inflation) − 1   (Fisher equation).
The first observation (Jul 1926) is dropped because monthly inflation
requires a prior CPI reading.
```

[3] Rolling 10-year window math — $1,000 lump sum

```
For each month t, compound the next N*12 monthly returns:
  terminal(t) = $1,000 * exp( Σ log(1 + r_i) )   for i in [t, t+N*12-1]
  CAGR(t)     = (terminal(t) / $1,000) ^ (1 / N) - 1
Log-returns are summed (not products chained) for numerical stability
over long horizons. Each window is labeled by its START month, not its
end. Windows lacking a full N*12 months are dropped from the tail.
```

[4] Full-history market CAGR

```
terminal = exp( Σ log(1 + r_t) ) over the full sample of monthly returns.
CAGR     = terminal ^ (12 / n_months) − 1.
Shown on the cover for context; independent of the scenario schedule.
```

Reproducibility

```
python -m src.ingest      # rebuild data/processed/monthly_returns.parquet
python -m src.export      # rebuild docs/data/*.json
python -m src.report      # rebuild docs/reports/*.pdf
Sample window: Aug 1926 – Mar 2026  (1,195 monthly observations).
```